



DATA SCIENCE COURSE PROJECT PROPOSAL

AI-Driven Water Quality Prediction and Explainable Decision Support System

A Data Science approach to predicting water quality and explaining the factors behind each prediction.

GROUP MEMBERS 05

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DEPARTMENT

Computer Engineering

ACADEMIC YEAR

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Water Quality Monitoring

Traditional water quality assessment relies on periodic sampling and laboratory analysis. This makes continuous assessment and early identification of deterioration difficult.

With increasing availability of water-quality datasets, Data Science and Machine Learning can be used to:



Analyze Patterns

Uncover trends and relationships across water-quality parameters.



Predict Conditions

Estimate current or upcoming water-quality status from data.



Identify Key Parameters

Determine which factors influence quality most strongly.



Support Decisions

Enable faster, evidence-based decision-making.

Existing Approaches



ML-Based Prediction

- Random Forest
- XGBoost
- SVM
- Decision Trees
- Regression models



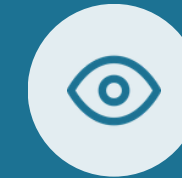
Deep Learning

- ANN
- CNN
- LSTM
- GRU



IoT-Based Monitoring

- Sensor-based data collection
- Real-time monitoring
- Cloud / edge-based systems



Explainable & Intelligent Systems

- Explainable AI
- Anomaly detection
- Decision-support systems
- Pollution prediction

Note: Our literature survey covered recent studies from 2022–2025 focusing on machine learning, IoT, WQI prediction, and intelligent water-quality monitoring.

What We Learned from the Literature

EXISTING WORK

Prediction

- Water quality classification
- WQI prediction
- Individual parameter prediction

Monitoring

- IoT sensors
- Real-time data collection
- Dashboards

Advanced Methods

- Deep learning
- Ensemble models
- Explainable AI



Common Challenges

- Limited datasets
- Sensor / data quality issues
- Poor model generalization
- Limited interpretability
- Lack of standardized datasets



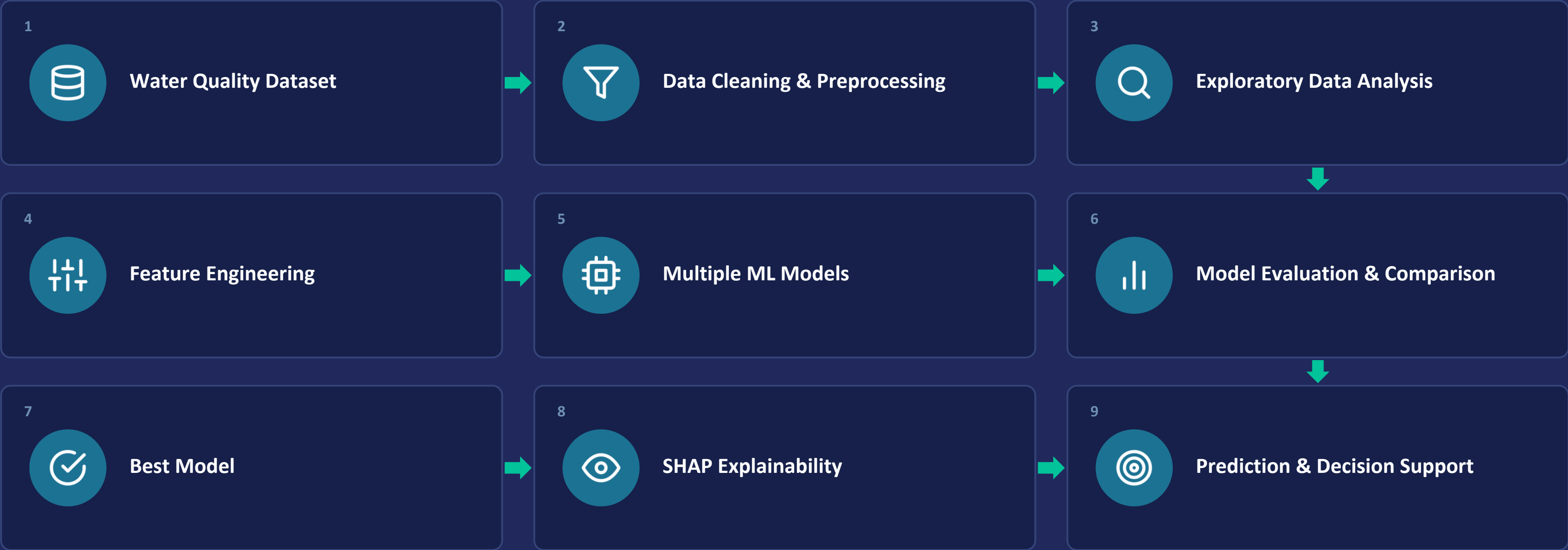
Opportunity for Our Project

Develop an end-to-end Data Science workflow combining:

**Prediction + Model Comparison + Explainability +
Decision Support**

Proposed Solution

The system will predict water quality from relevant parameters and provide an interpretable explanation of the factors influencing each prediction.



Methodology



Data Analysis

- Missing values
- Outlier detection
- Correlation analysis
- Feature distributions
- Feature engineering



Machine Learning

Models compared:

- Logistic Regression
- Decision Tree
- Random Forest
- SVM
- XGBoost
- KNN

Selection depends on whether the final dataset supports classification or regression.



Evaluation

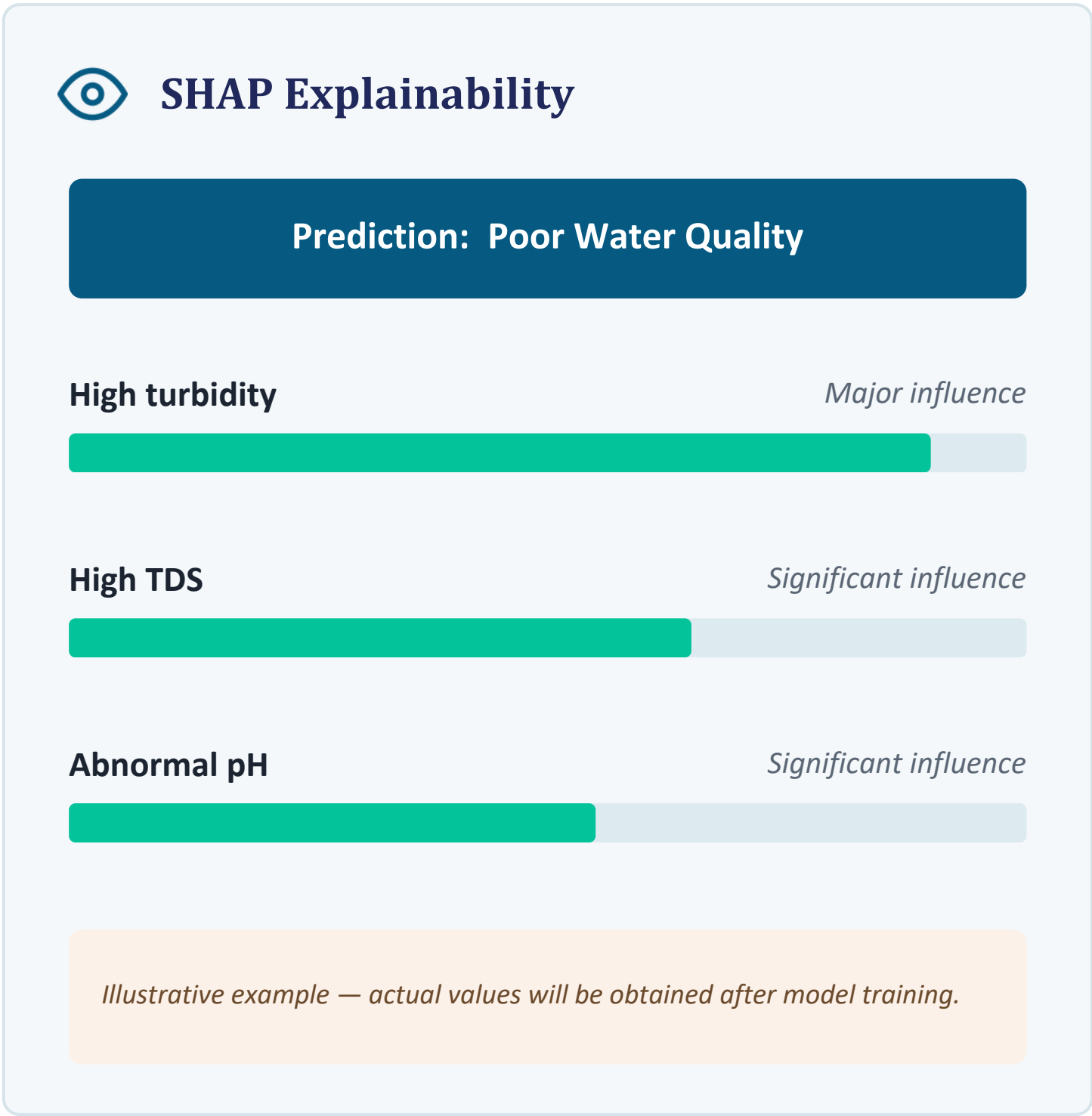
Classification

- Accuracy
- Precision / Recall
- F1-score
- Confusion Matrix

Regression / WQI

- MAE
- RMSE
- R^2

Beyond Prediction



EXPECTED OUTCOME & SCOPE

Expected Deliverables

1

Cleaned & analyzed water-quality dataset

2

Exploratory Data Analysis

3

Multiple ML models

4

Comparative performance evaluation

5

Best-model selection

6

SHAP-based explainability

7

Interactive Streamlit dashboard

8

Water-quality decision support

FINAL OBJECTIVE

To develop an end-to-end Data Science application that transforms water-quality data into predictions, interpretable insights, and actionable decision support.

Proposed Project → **Water Quality Prediction + Explainable AI + Decision Support**